CS-80 Reference: Improvement Plan and Status

How the emulation will match the original Yamaha CS-80 in sound, features, function and UI/UX, and go beyond Arturia CS-80 V4. The report shows what is implemented and exactly what remains.

Repository `cheshmakzanoon-ops/yamaha-cs80` · branch `claude/clever-babbage-sm9ifa` · state as of 31 October 2026, head commit `7016634`

At a glance. The plan has eight phases (0–7). So far:

- **Phase 0 (foundations):** its safe parts are done.
- **Phase 1 (the voice):** three of its seven steps are done: gain staging, RC envelopes, and the first IG00156 filter stage.
- **Phases 2–7:** not started.

Every completed step was built and tested locally. The tests are all core tests plus the plugin UI self-test, and the legacy engine stays byte-identical. Each step is pushed with its own commit.

1. The goal and the decisions behind it

Goal. Every sound, feature, function and UI element should match the original CS-80 as far as public evidence and sound engineering allow. The result should also be demonstrably better than Arturia CS-80 V4.

Decision (taken with the owner)	What it means in practice
Evidence: public sources only	The models use Yamaha schematics, the service manual, the adjustment procedures, and declared engineering inference. There is no hardware-capture phase. Public recordings are used only for blind listening, never as calibration.
Engine: circuit engine becomes the default	New instances and presets will use the circuit-modelled engine (today called "Source Performance"). Legacy stays only so that old sessions load and sound exactly as before.
UI: the original panel comes first	The main view reproduces the CS-80 panel arrangement. Modern extras (signal-flow plate, library, modulation pages) move into a slim toolbar, drawers or a second "Studio" view.
Modern layer: match and exceed Arturia	Add drawable envelopes (MSEGs), a mod matrix, MTS-ESP/Scala microtuning, MIDI learn, undo/redo, A/B compare and better effects. All of these are inactive in a "Pure CS-80" mode.

How "better than Arturia" is made demonstrable

1. A traceable engine: every law cites a schematic page or a declared, labelled inference.
2. Circuit behaviours modelled explicitly: free-running voices, KAS scanning and sample/hold, and the PRA wah, OE1/OE2 chorus/tremolo and output networks.
3. An in-app evidence view showing which behaviour is source-derived and which is inferred.
4. Offline renders at a higher quality than realtime.
5. A "Pure CS-80" switch that removes every non-original feature.
6. A modern layer at least equal to Arturia V4's, plus better ergonomics.

The Arturia V4 feature list was taken from search results quoting Arturia's product page, retailer listings and the V4.0.1 manual. arturia.com itself was not reachable from the build environment.

Guiding architecture: truth model vs realtime model

With no hardware to measure, each stage gets an offline **truth model** built directly from the schematic equations. A truth model uses double precision, high oversampling and implicit or Newton solves. The realtime model is then held to it by tolerances fixed before testing. The repository already contains this pattern in `src/reference/*`. Every intended change to the circuit engine's sound is documented, and the CI audio gate is re-pinned to the commit that makes it. Legacy audio must never change.

2. Starting point (what the audit found)

Area	Finding at the start of this work
Filter	A linear TPT state-variable filter with a soft clip on its input, outside the resonance loop. That clip was the main source of aliasing: saw spurs reached −34 to −42 dB.
Oscillator	PolyBLEP, with a placeholder 3.5 % third harmonic for the sine shaper and a fixed 30 µs reset pulse. A circuit-faithful oscillator existed but was not active.
Envelopes	An `exp(−6/…)` law, which reaches only −52 dB although the comments say −60 dB. A ×5 "bend" was applied in fixed time windows, and the filter envelope reset to its initial level when a voice was reused.
VCA, smoothing, variation	The VCA is a linear multiply and no parameter is smoothed. Drift is a single 0.031 Hz sine of about ±22 cents, and 3 of the 5 voice trims are never used.
Engine default	The circuit engine is off by default, and all 222 library programs force Legacy, so most users never heard the circuit work.
Levels and CPU	Circuit tone tabs reached +3.2 to +3.9 dBFS. About 13 transcendental calls per voice-rank per sample, at a fixed 2× oversampling.
UI/UX	A strong look, but some placements differ from the original. There was no undo, MIDI learn, A/B, typed values, right-click menu, fine-drag or computer-keyboard play, and the engine-mode switch was buried in Advanced → page 4.
Product	There are 178 randomly generated library programs, no Program Change, no host parameter groups, no LICENSE file, no signing or installers, and an intermittent Linux pluginval crash.

3. What has been implemented

Commit	Step	What it did	Measured result
`97cc815`	Plan recorded	Saved the full plan as `docs/IMPROVEMENT_PLAN.md`.	—
`7b55d34`	**DONE** 0.1 Exact speed-ups	Filter, envelope, key-voltage, cutoff and glide maths is skipped while its inputs are unchanged. The PWM sine is computed once per rank instead of per voice, the decimator no longer branches on every tap, and dead code was removed. Output is bit-identical.	Circuit engine 0.298→0.230 RTF Legacy 0.274→0.248 RTF
`eb486cb`	**DONE** 0.3 Save-state version **DONE** 0.5 Status page	Sessions and presets now record `stateSchemaVersion`. Old saves keep Legacy even after the default flips. `docs/STATUS.md` is the single current-state page.	UI self-test passes
`228459c`	**DONE** 1.5 Gain staging	A fixed −6 dB output reference (a software calibration), plus an inferred smooth output-driver ceiling at full scale.	Tab peaks +3.2→−2.9 dBFS 0 samples over full scale

`131f72b`	**DONE** 1.3 RC envelopes	Decay and release are now true time-to−60 dB curves. The attack charges toward an overshoot target and a comparator stops it at the peak, reached at exactly T. The ×5 bends are gone, and the filter envelope continues instead of jumping when a voice is reused.	Worst steal click 1.17→0.90 Retrigger 1.06→0.96
`f6b126d`	**FIRST STAGE** 1.1 IG00156 filter	Saturation now sits inside the filter loop (an OTA tanh law, solved per sample by Newton iteration), and the input clip is removed. The 4.7 MΩ feedforward paths from the schematic add a 22.6 Hz floor to each cutoff.	Saw aliasing −33.8→−51.1 dB Pulse −39.3→−53.5 dB CPU 0.23→0.37 RTF

`cbb0de1`
`e8b163a`
`7016634`
	CI re-pins	After each intentional sound change, the circuit-engine audio gate was moved to that commit. The older commits stay as references for earlier renders.	DSP checks passed on `e8b163a`

Known side effects, documented.

- Some low-cutoff string and brass tabs are now 5–6 dB louder and brighter. This comes from the 22.6 Hz floor raising very low cutoffs, which is a consequence of the documented circuit.
- An extreme 8-note velocity-127 chord peaks at 1.017, about 0.15 dB over full scale. The driver stops at 1.0, and the output coupling capacitor after it adds a small transient.
- CI flagged the known intermittent Linux pluginval GUI crash once, on `7b55d34`. The next build passed.

Items moved or deferred from Phase 0, and why

Item	Status and reason
Parameter smoothing	**MOVED TO PHASE 1** Both pinned CI audio probes move controls during playback, so smoothing changes their output. It is an audible change, and it will be applied to the circuit engine only.
Splitting the 420-line sample function; fast maths approximations; structure-of-arrays layout	**MOVED TO PHASE 1** These change the circuit engine's pinned audio bytes, so they ship together with the Phase 1 sound changes.
Shared test helpers (`tests/TestSupport.h`)	**DEFERRED** They will be written when the first new stage gate needs them.
Moving historical docs into `docs/evidence` and `docs/history`	**NOT DONE** Skipped to avoid churn and broken links. `STATUS.md` already removes the conflicting status statements.

4. What remains: exactly what needs to be done

Order of execution: finish Phase 1 in the order below, then Phase 5.1–5.4 alongside Phase 2, then Phases 2, 3 and 4, then 6, then 7. Every step needs passing core tests, the UI self-test and pluginval in CI, and byte-identical legacy audio. Every step also needs a documented and re-pinned circuit-engine change, plus `STATUS.md` and coverage-ledger updates.

Phase 1: the voice (remaining)

Step	Exactly what to do	Done when
TODO 1.1b Complete the IG00156 nodal model	• Add the 1.2 MΩ/180 pF feedback branch, which is the third state of `DynamicModel`. • Add a declared hypothesis for the Q-block VQ→conductance law, meeting Q 5 at 0 V and Q 0.5 at 10 V. • Model the separate HP and LP input and ground networks, and the 1 µF interstage coupling. • Verify against `src/reference/IG00156Linear.h` in small signal.	Small-signal error is within the pre-registered tolerance over the cutoff × Q grid, and the roadmap W1 anchors are reproduced.
TODO 1.2 Oscillator promotion	• Move `src/experimental/ReconstructedOscillator.h` into the core and use it for the circuit engine. • Re-qualify it against the new filter: it failed only against the old clipping filter. • Share its 22 KB residual table across instances. • Replace the 3.5 % third harmonic with its piecewise triangle-to-sine shaper.	Alias at or below −70 dB in the 48 kHz alias suite. Pin-level anchors hold (saw 3.5 Vpp, sine 3 Vpp). CPU is within budget.
TODO 1.4 VCA (IG00151)	Separate the exponential and linear control inputs. Add a small CV feedthrough and offset as declared inference, and use a constant-power rank pan.	Level-law tests pass, and the steal and retrigger click scores stay at or below 1.
TODO 1.6 Oversampling quality	• Add a new appended parameter with three settings: Eco (2×), Standard (the lowest factor that meets the alias target within budget) and HQ (8×). • Offline bounces always use HQ, detected with `isNonRealtime()`. • Compute the decimator and reported latency per mode.	Latency is reported correctly per mode, and pluginval passes.
TODO Smoothing and engine restructure	• Add sample-rate-independent smoothers for every continuous control and controller, on the circuit engine only. • Split `processInternalSample` into global, rank, mix and output parts. • Use fast bounded-error `exp2` / `tan` / `pow`, and a structure-of-arrays layout across the 16 rank chains.	No zipper noise under automation. Circuit engine at or below 0.15–0.20 RTF for 8 voices at 48 kHz, and idle at or below 2 %.
TODO 1.7 Engine default flip	• Fresh instances and new presets use the circuit engine; schema-1 states stay Legacy. • Replace the two cryptic toggles with **ENGINE: Circuit | Legacy (compat)** and **PURE CS-80**. Host IDs stay unchanged.	Old-session and preset recall tests pass, and new instances sound the circuit engine.

Phase 2: voice behaviour, keyboard and variation

Step	Exactly what to do	Done when
TODO 2.1 Free-running voices	• Wire `ContinuousVoiceCircuit` into the engine: held key voltage, oscillators that run without a note-on reset, and filter state that keeps evolving. • Add a cheap idle path, such as decimated idle evolution with an exact catch-up when the voice wakes.	The `VOICE_CIRCUIT_CONTINUITY.md` criteria are met within the CPU budget.
TODO 2.2 KAS scan and sample/hold	Model scan timing and priority, the per-voice held-key-voltage droop (0.033 µF node) and the NAND(TR, SUS) hold. Choose a declared policy for whether a 9th held key is reconsidered.	Allocation regressions and droop sensitivity tests pass.
TODO 2.3 Keyboard contour key map	Choose a declared KAS/KBC mapping and keep its alternatives listed. Drive audio from `src/reference/KeyboardContour.h`, and delete the provisional weights in `PerformanceControls.h`.	All 61 keys are tested against the lever grid, and the C2/C5# anchor conflict is reported openly.
TODO 2.4 Touch	Model the TWS b/C acquisition (held initial touch, AC-coupled pitch transient) and calibrate the TRG5 pressure bus. Add velocity and pressure curve adapters for different MIDI controllers.	The `INITIAL_TOUCH_CIRCUIT_EVIDENCE.md` gate list passes.
TODO 2.5 Tolerances and drift	• Add per-card component tolerances (declared 5 % resistor and 10 % capacitor distributions), mapped into tune, cutoff, Q, level and envelope trims. • Replace the sine drift with thermal 1/f drift and a warm-up curve. • Keep today's ±22 cents as a "worn unit" preset.	All five trims are wired and the distributions are documented.

Phase 3: modulation, effects, output

Step	Exactly what to do	Done when
TODO SUB (IG00150)	An exponential speed converter, including touch currents, anchored at 200 Hz, with band-limited waveforms where the aliasing would be audible.	The speed law matches its anchor and the waveforms are alias-checked.
TODO Ring modulator	An envelope-to-frequency law from the ADJ p19 anchors (150 ms and 5 ms periods), plus multiplier DC offset and carrier feedthrough. Mask the 500 Hz extension in PURE mode.	The anchors are reproduced and sidebands tested.
TODO OE1/OE2 chorus and tremolo	Model tremolo from OE1. In the BBD, add per-stage charge loss, clock feedthrough, noise and true left/right wet paths. Use `Mn3001Packet.h` and `Oe2Filters.h` as truth models.	The stereo transfer is tested against the truth models.
TODO PRA wah and EXP pedal	An optical cell with asymmetric attack and decay (a declared LDR model).	Sweep dynamics are tested.

TODO Output and noise	Make the GENERAL mono jack selectable and add an optional headphone-amp model. Shape the noise spectrum, and decide which noise sources are shared and which are per voice.	The output modes are tested and the noise statistics documented.

Phase 4: tones, memories and library

Step	Exactly what to do	Done when
TODO 22 tone tabs	Re-read the 25 unknown matrix cells from native-resolution crops, and keep any that stay unresolved as labelled hypotheses. Refine the BA driver impedance and diode I-V.	Each cell is either transcribed or explicitly marked as a hypothesis.
TODO Memory panels	Model the electrical loading and switch states for the four memories and PANEL.	Switch and loading tests pass.
TODO Curated factory library	About 300 hand-voiced programs on the circuit engine, categorised and tagged, replacing the 178 random variants. The old 222 programs move to a read-only Legacy Library.	Old sessions still recall the legacy bank.
TODO MIDI	Program Change, and a MIDI CC for the ribbon.	Host tests pass.

Phase 5: UI/UX, original panel first

Step	Exactly what to do
TODO 5.1 Panel conformance audit	Write `docs/PANEL_CONFORMANCE.md`: a control-by-control comparison against owner's-manual index items 1–46 and the service-manual S2 panel drawing. Cover position, group, legend, cap colour, lever direction and switch type, and record conflicts.
TODO 5.2 Re-lay the main view	• Remove the signal-flow plate so the channel rows span the panel. • Move the foot and contour groups to the side the evidence shows. • Use separate chorus and tremolo switches. • Print full tone-tab names. • Match the memory-lid placement. • Replace the ±24 px layout hack with one layout table.
TODO 5.3 Tuck away modern items	A slim toolbar (preset, library, A/B, undo/redo, ENGINE/PURE, menu). A "Studio" view with Modulation, Keyboard/MPE/Tuning, FX, Arp and Settings tabs. An optional scope drawer. The pitch and mod wheels go in a modern strip that is hidden in PURE mode.
TODO 5.4 Interaction polish	• Faders: relative drag with no jump, Shift for fine adjustment, Ctrl/Cmd-click to type a value, double-click to reset. • A right-click menu with MIDI learn, reset, type value and copy/paste. • A full-range ribbon from any touch point. • Computer-keyboard play with octave shift. • Library: audition and prev/next. • Memory lid: store and copy. • Plain-language tooltips with "Circuit notes".
TODO 5.5 Readability and rendering	Embed an OFL condensed typeface, with a 10.5 px legend floor and a WCAG contrast pass. Add zoom presets from 75 % to 200 %, and render the HiDPI crinkle texture at physical scale.

TODO 5.6 Editor code health	• Create one `ArtCommon.h` to replace the three duplicated helper sets, and delete dead code. • Split the ~1,830-line editor implementation into Toolbar, ChannelRow, CentreRow, PerformanceDeck and MemoryLid. • Cache the memory-lid art, make the cap variant deterministic, and use an LRU paint cache. • Replace the 8 Hz parameter polling with listeners.
TODO 5.7 UI tests	Golden-image diffs at 1600×1000 and 1120×700, plus tab-order, minimum hit-target, library-behaviour and context-menu tests, and an accessibility review.

Phase 6: a modern layer that beats Arturia V4 (all inactive in PURE CS-80)

Area	Exactly what to build
TODO Modulation	3 drawable, loopable, tempo-syncable MSEGs. LFO waveforms beyond sine (today it is hard-coded to sine). A 16-slot modulation matrix whose sources include ribbon, pressure, contour, SUB and the ring envelope. A keyboard modulation page.
TODO Tuning and MIDI	MTS-ESP and Scala/KBM microtuning. MPE with RPN zone negotiation, an upper zone and high-resolution velocity. MIDI learn and Program Change.
TODO Editing	Undo/redo through the APVTS UndoManager, A/B compare, copy rank I↔II and memory↔panel, and "Randomise within original ranges".
TODO Host and engine	Host parameter groups, which already exist in the code but are unused. Voice modes: 8 authentic voices, plus optional 16-voice and unison modes. Optionally, sample-accurate automation.
TODO Effects	Gain-compensated FX, better plate and tape models, a BBD ensemble, a spring reverb, and per-slot oversampling for the drive.
TODO Learning	A tutorial and tour overlay. Per-control help that cites the owner's manual page and shows the evidence class.

Phase 7: release qualification

- TODO Root-cause the intermittent Linux pluginval exit 139 instead of relying on reruns.
- TODO Run real-DAW sessions: Live, Logic, Reaper, Bitwig, Cubase and FL Studio.
- TODO Choose a project LICENSE and settle JUCE licensing (AGPL or commercial). Get a legal review of the "CS-80" and "Yamaha" names in the product name, bundle ID and nameplate art.
- TODO Sign and notarise, build installers and a universal macOS binary. AAX only with Avid SDK and PACE agreements.

5. How every step is verified

- Core build and `ctest` (44 suites today), plus the plugin build and the UI/host self-test through `scripts/run-ui-smoke-linux.sh`.
- Legacy byte-identity is checked locally against the pinned probes and in CI. The circuit-engine audio gate is re-pinned only for documented, intentional changes.
- Truth-model vs realtime residuals use tolerances fixed before evaluation. Every new test is mutation-checked: the old behaviour is deliberately put back to confirm the test fails.
- `cs80_engine_audit` covers alias, clicks, headroom and loudness, at 44.1, 48 and 96 kHz.
- CPU is measured on a benchmark at 44.1, 48 and 96 kHz, and pluginval strictness 10 runs on all four platforms in CI.
- Screenshots are refreshed at 1600×1000 and 1120×700. Blind, level-matched listening is labelled as listening evidence, not calibration.

6. Scorecard

Metric	Start	Now	Target
Worst saw aliasing, circuit engine, 48 kHz	−33.8 dB	−51.1 dB	≤ −70 dB (needs Phase 1.2)
Worst tone-tab peak, circuit engine	+3.2 dBFS	−0.8 dBFS	≤ 0 dBFS
Worst voice-steal click score	1.17	0.93	≤ 1.0
Circuit-engine CPU, 8 voices, 48 kHz	0.30 RTF	0.37 RTF	≤ 0.15–0.20 RTF
Plan steps complete	0	Phase 0 (safe parts) + 3 of 7 Phase 1 steps	All of Phases 0–7

The CPU figures come from a scratch benchmark of 8 held notes at 48 kHz with 256-sample blocks, on a shared cloud core. Treat them as relative, not absolute. The alias and click figures come from `cs80_engine_audit` and are software measurements, not comparisons with a hardware CS-80.